The Free ℓ^1 Seminorm on Banach Presheaf Coboundaries

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Abstract

We classify seminorms on the 1-cochain space of a Banach presheaf over a finite poset satisfying additivity, faithfulness, and functoriality with respect to bounded linear maps (contractive monotonicity). Specifically, let P be a finite partially ordered set and F a Banach presheaf over P. The **1-cochain space** is $C^1(P, F) = \bigoplus_{e \in E_{\mathrm{cov}}} F(a_e)$, where the direct sum is taken in the ℓ^1 sense. The space $C^1(P, F)$ is the free additive Banach object generated by the edges. We prove that the ℓ^1 coboundary norm $I(s) = \sum_e \|\delta_e(s)\|$ is, up to a positive scalar, the **unique edge-additive seminorm** on $C^1(P, F)$ satisfying these three properties. Adding isomorphism invariance forces the scalar to be uniform across edges. No additional choice remains within this class: the ℓ^1 norm is uniquely determined by the axioms, which encode compatibility with the underlying Banach geometry.

In short: *any additive, faithful diagnostic on Banach presheaf coboundaries that is functorial under arbitrary bounded linear maps must be a positive scalar multiple of the ℓ^1 norm.*

Conventions. Norms are denoted $\|\cdot\|$ with appropriate subscripts; scalar absolute values are denoted $|\cdot|$.

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1. Setup

1.1 Banach Presheaves over Finite Posets

Let $(P, \leq)$ be a finite partially ordered set. The **covering relations** are the edges of the Hasse diagram:

$$E_{\mathrm{cov}}(P) = \{e = (a, b) : a < b,\ \nexists\, c \text{ with } a < c < b\}.$$

A **Banach presheaf** F over P assigns: - To each $x \in P$, a Banach space $(F(x), \|\cdot\|_{F(x)})$. - To each $e = (a, b) \in E_{\mathrm{cov}}$, a bounded linear **restriction map** $r_{b,a} : F(b) \to F(a)$ with $\|r_{b,a}\|_{\mathrm{op}} \leq 1$.

A **section** of F is a family $s = (s(x))_{x \in P}$ with $s(x) \in F(x)$.

1.2 The 1-Cochain Space

For each edge $e = (a, b)$, the **coboundary defect** is:

$$\delta_e(s) = r_{b,a}(s(b)) - s(a) \in F(a).$$

The **1-cochain space** is the Banach direct sum:

$$C^1(P, F) = \bigoplus_{e \in E_{\mathrm{cov}}}^{\ell^1} F(a_e)$$

with norm $\|(\delta_e)_e\|_1 = \sum_e \|\delta_e\|_{F(a_e)}$. This is isometrically isomorphic to the space of coboundary vectors. As discussed in Section 6.2, $C^1(P, F)$ acts as the free additive Banach object generated by the edges.

The ℓ^1 coboundary norm of a section s is:

$$I(s) = \sum_{e \in E_{\mathrm{cov}}} \|\delta_e(s)\|_{F(a_e)} = \|(\delta_e(s))_e\|_1.$$

1.3 Seminorms on 1-Cochains

> **Definition 1.1.** A **seminorm on 1-cochains** is a function $\nu : C^1(P, F) \to \mathbb{R}_{\geq 0}$ satisfying $\nu(cx) = |c|\nu(x)$ and $\nu(x+y) \leq \nu(x)+\nu(y)$ for all $x, y \in C^1(P, F)$ and scalars c.

We consider four properties:

(P1) Edge-wise additivity. ν decomposes as

$$\nu((\delta_e)_e) = \sum_{e \in E_{\mathrm{cov}}} \nu_e(\delta_e)$$

where each $\nu_e : F(a_e) \to \mathbb{R}_{\geq 0}$ is itself a seminorm on $F(a_e)$.

(P2) Faithfulness. For each e: $\nu_e(\delta) = 0 \iff \delta = 0$.

(P3) Functoriality under bounded linear maps (Contractive Monotonicity). The function ν_e is compatible with the underlying Banach geometry. Specifically, for every bounded linear endomorphism $\varphi : F(a_e) \to F(a_e)$ on the stalk space:

$$\nu_e(\varphi(\delta)) \leq \|\varphi\|_{\mathrm{op}} \cdot \nu_e(\delta).$$

(This condition is intentionally strong: functoriality under all bounded linear endomorphisms forces the seminorm to depend only on the underlying Banach norm, and is the key structural input driving the classification.)

(P4) Isomorphism invariance. For any poset isomorphism $\pi : \tilde{P} \to P$:

$$\nu_{\tilde{P}}(\pi^*(\delta_e)_e) = \nu_P((\delta_e)_e).$$

1.4 Relation to the Combinatorial Classification

The present classification may be viewed as a functional-analytic extension of the finite-dimensional result established in "Coordinate-Wise Additivity and the ℓ^1 Norm on Finite Graph Cochains" [4]. In that setting, ℓ^1 arises from local additivity over disjoint coordinate supports and edge separability alone, with no Banach space structure required. Here, we replace coordinate additivity with structural compatibility in the form of functoriality under bounded linear maps on Banach spaces (P3), showing that the same rigidity persists in a substantially more general setting. The combinatorial result [4] thus provides the minimal case; the present paper demonstrates that the ℓ^1 conclusion is not an artifact of finite coordinates, but survives the passage to infinite-dimensional Banach stalks.

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2. The Classification Theorem

We first isolate two structural lemmas.

Lemma 2.0 (1-Dimensional Compatibility). Let ν_e satisfy (P3). Then for any 1-dimensional Banach subspace $\mathrm{span}(\delta) \subset V$, the restriction $\nu_e|_{\mathrm{span}(\delta)}$ is well-defined and satisfies (P1)–(P3) with respect to the induced Banach norm on $\mathrm{span}(\delta) \cong \mathbb{R}$. Specifically: any isometric embedding $\iota : \mathbb{R} \hookrightarrow V$ induces a seminorm $\nu_e^{\mathbb{R}} = \nu_e \circ \iota$ on $\mathbb{R}$ satisfying (P2)–(P3), and any bounded linear functional $f : V \to \mathbb{R}$ induces $\nu_e^{\mathbb{R}}(f(\delta)) \leq \|f\|_{\mathrm{op}} \cdot \nu_e(\delta)$.

Proof. Direct application of (P3) to ι and f respectively. Both are bounded linear maps, so (P3) applies without additional hypothesis. □

Lemma 2.1 (Direction Independence). If ν_e satisfies (P3), then for any $v, w \in F(a_e)$ with $\|v\| = \|w\| \neq 0$, we have $\nu_e(v) = \nu_e(w)$.

Proof. By the Hahn–Banach theorem, there exists $f_v \in F(a_e)^*$ with $\|f_v\|_{\mathrm{op}} = 1$ and $f_v(v) = \|v\|$. Define $\Phi : F(a_e) \to F(a_e)$ by $\Phi(x) = f_v(x)\frac{w}{\|w\|}$. Because Φ is a bounded linear endomorphism of $F(a_e)$, (P3) applies directly here. Then $\Phi(v) = w$ and $\|\Phi\|_{\mathrm{op}} = 1$, so $\nu_e(w) \leq \nu_e(v)$ by (P3). By symmetry, constructing $\Psi(x) = f_w(x)\frac{v}{\|v\|}$ gives $\nu_e(v) \leq \nu_e(w)$, forcing equality. □

Theorem 2.2 (Free ℓ^1 Seminorm Classification). Under functoriality with respect to all bounded linear endomorphisms, let ν be a seminorm on $C^1(P, F)$ satisfying (P1), (P2), and (P3). Then for each edge e, there exists $w_e > 0$ such that

$$\nu_e(\delta) = w_e \cdot \|\delta\|_{F(a_e)} \quad \text{for all } \delta \in F(a_e).$$

If additionally (P4) holds, then $w_e = \alpha$ is independent of e, and

$$\nu((\delta_e)_e) = \alpha \sum_e \|\delta_e\|_{F(a_e)} = \alpha \cdot I(s).$$

In other words: the ℓ^1 coboundary norm is the unique (up to positive scalar) edge-additive seminorm satisfying all four properties.

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3. Proof

The proof proceeds in three stages.

Stage 1: Direction Independence

This is Lemma 2.1: for any two vectors $v, w \in F(a_e)$ with $\|v\| = \|w\| \neq 0$, we have $\nu_e(v) = \nu_e(w)$.

Stage 2: Each ν_e is proportional to $\|\cdot\|$.

Fix an edge e and let $V = F(a_e)$. By Lemma 2.1, all unit vectors in V have the same ν_e-value: there exists $w_e \geq 0$ such that $\nu_e(v) = w_e$ for all v with $\|v\| = 1$. By (P2), $w_e > 0$.

For any $\delta \in V$ with $\delta \neq 0$, the absolute homogeneity of ν_e gives:

$$\nu_e(\delta) = \|\delta\| \cdot \nu_e\left(\frac{\delta}{\|\delta\|}\right) = \|\delta\| \cdot w_e = w_e\|\delta\|.$$

Since $\nu_e(0) = 0$ by the seminorm property, we conclude $\nu_e(\delta) = w_e\|\delta\|$ for all $\delta \in V$. $\square$ (Stage 2)

Stage 3: Isomorphism invariance forces uniform weights.

Because of (P1), the seminorm **splits as a direct sum**, so each component can be evaluated in isolation. Let $e = (a, b)$ be any covering edge of a finite poset P. Define the single-edge subposet $P_e = \{a < b\}$. Each summand $\nu_e(\delta_e)$ depends only on the coboundary defect $\delta_e(s) = r_{b,a}(s(b)) - s(a)$, which is computed entirely from the restricted presheaf data $(F(a), F(b), r_{b,a})$. The value $\nu_e(\delta_e)$ is therefore intrinsically independent of the ambient poset P in which e is embedded: it is determined by the single-edge presheaf (P_e, F_e).

By Stages 1–2, $\nu_e(\delta_e) = w_e\|\delta_e\|$ for some $w_e > 0$. Since w_e depends only on the isomorphism class of the single-edge presheaf (P_e, F_e), it remains to show all such presheaves yield the same weight.

Let $e_1 = (a_1, b_1)$ and $e_2 = (a_2, b_2)$ be any two covering edges with the same stalk spaces $F(a_{e_1}) \cong F(a_{e_2})$ and $F(b_{e_1}) \cong F(b_{e_2})$, and compatible restriction maps

under the induced isomorphism. The single-edge posets $P_{e_1} = \{a_1 < b_1\}$ and $P_{e_2} = \{a_2 < b_2\}$ admit a unique order-preserving bijection $\pi : P_{e_1} \to P_{e_2}$. By (P4), $\nu_{P_{e_1}}(\pi^*(\delta_{e_2})) = \nu_{P_{e_2}}((\delta_{e_2}))$, which gives $w_{e_1} = w_{e_2}$.

Thus all covering edges share the same weight w. Setting $\alpha = w$: $\nu = \alpha \cdot I$. □ (Theorem 2.2)

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4. Necessity of Each Property

Dropped property	What happens	Example
(P1) Additivity	ℓ^∞-type norms admitted	$\nu = \max_e \|\delta_e\|$
(P2) Faithfulness	Trivial seminorm admitted	$\nu \equiv 0$
(P3) Functoriality	Direction-dependent seminorms admitted	On $F(a_e) = \mathbb{R}^2$: $\nu_e(x_1, x_2) = \|x_1\| + 2\|x_2\|$; satisfies (P1), (P2), (P4), but violates direction independence (Lemma 2.1), hence is not proportional to $\|\cdot\|$
(P4) Invariance	Non-uniform weights admitted	$\nu = \sum_e w_e \|\delta_e\|$, w_e edge-dependent

Each property is independently necessary. Together they are sufficient.

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5. Structural Uniqueness

Rather than defining artificial categories to encapsulate simple scaling relations, we formulate the universal structure directly:

> **Proposition 5.1 (Normalized Uniqueness).** The ℓ^1 coboundary norm is the unique normalized additive diagnostic satisfying properties (P1)–(P4). Any evaluation compatible with these constraints is identically $\alpha \cdot I(s)$ for some positive scalar α.
>
> **Corollary 5.2 (Commutative Factorization).** For any seminorm ν satisfying (P1)–(P4) and any bounded linear map φ acting stalkwise, the evaluation functional factors predictably:

$$\nu(\varphi_*(\delta)) \leq \|\varphi\|_{\mathrm{op}} \cdot \nu(\delta),$$

and the ℓ^1 coboundary norm I is the unique (up to scalar) edge-additive framework for which this structural compatibility with bounded linear maps identically holds.

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6. Discussion

6.1 What This Paper Proves

This paper isolates a purely structural constraint on how coboundary defects in Banach presheaf systems can be measured: once one requires edge-wise additivity, faithfulness, and compatibility with the ambient Banach geometry through functoriality under bounded linear maps, the resulting seminorm is forced to be ℓ^1 up to a global scalar. The classification does not depend on any dynamical assumptions, probabilistic structure, or physical interpretation; it is a consequence of the algebraic and geometric properties encoded in (P1)–(P4). The uniqueness result is therefore contingent on allowing the full algebra of bounded linear maps; restricting to a smaller class of transformations would admit additional seminorms. In particular, the result shows that the freedom to choose a defect metric on the 1-cochain space is more limited than it may appear — alternative geometries such as ℓ^2 or ℓ^∞ are not compatible under these full constraints. The significance of the theorem is therefore not the introduction of a new norm, but the identification of conditions under which no alternative exists.

6.2 Relation to Other Work

The Hahn–Banach factoring argument used in Stage 1 is analogous to classical norm classification techniques in Banach space theory, where linear functionals are used to reduce geometric constraints to one-dimensional analysis (see Megginson 1998 [2] for Banach geometric tools, and Pestov 2006 [1] for related uniqueness results for norms). The key structural observation is that functoriality under all bounded linear endomorphisms, combined with Hahn–Banach, forces direction independence. This mechanism parallels the uniqueness proof for the operator norm.

The result is a presheaf version of a classical principle in Banach space theory: the characterization of the ℓ^1 norm as the unique additive cross-norm on a free Banach sum. The 1-cochain space $C^1(P, F) = \bigoplus_e F(a_e)$ is the *free additive Banach object generated by the edges* of the poset. When a measurement on such a free object must satisfy piecewise additivity and strict compatibility with the underlying norms on each component, ℓ^1 is the only geometry that survives. The ℓ^2 norm fails because $\|(v, w)\|_2 \neq \|v\| + \|w\|$; the ℓ^∞ norm fails because $\max(\|v\|, \|w\|)$ discards summation. This positions the ℓ^1 coboundary norm as the canonical additive cross-norm on the free Banach object $C^1(P, F)$ under the stated constraints.

We note that the coboundary defects $\delta_e(s) = r_{b,a}(s(b)) - s(a)$ measure the failure of local data to glue consistently across the covering relations of the poset. The distinction in our framework is that the choice of how to measure these defects is not arbitrary: it is uniquely forced to be the ℓ^1 coboundary norm by the structural axioms (P1)–(P4).

6.3 Strategic Place in the Overall Program

This paper sits as Step 2 within a four-paper sequence establishing that additive, structure-preserving measurements of local defects are uniquely forced to be ℓ^1 across progressively richer structural settings, and that projection dynamics generically produce nonzero instances of such defects. The sequence progresses: 1. **Coordinate-Wise Additivity** [4]: Local additivity over disjoint coordinate supports on finite graph cochains already forces ℓ^1 geometry. This is the purely combinatorial core. 2. **Free ℓ^1 Seminorm Classification** (this paper): The same ℓ^1 rigidity persists when the cochain spaces are replaced by Banach presheaves and coordinate additivity is replaced by structural compatibility (functoriality under bounded linear maps). 3. **Single Obstruction Theory** [3]: Dynamical retractions produce intrinsic coboundary defects measured by the forced ℓ^1 diagnostic, and these defects persist with non-vanishing density over finite-time intervals protected by Lieb–Robinson bounds. 4. **Hodge Decomposition** [5]: The forced ℓ^1 geometry induces a canonical quotient norm on $H^1(G)$, providing the unique gauge-invariant measure of irreducible obstruction.

6.4 Connection to Dynamical Obstruction Theory

The classification result established here is purely structural: it identifies the unique form of any additive, functorial diagnostic on presheaf coboundaries. Its significance becomes more apparent when combined with dynamical settings in which such coboundary defects are generated.

In the companion work "Projection Obstruction Theory" [3], projection of linear many-body dynamics onto a nonlinear manifold is shown to produce intrinsic coboundary defects that cannot be removed by linearization and that persist with non-vanishing density over finite time intervals. The present result implies that any local, additive measurement of such defects must coincide with the ℓ^1 coboundary norm (up to scaling). Thus, the ℓ^1 diagnostic appearing in that analysis is not a modeling choice, but the only one compatible with the structural constraints imposed here.

Acknowledgments

No external funding. No conflicts of interest.

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